Prediction of Student Learning Style Using Modified Decision Tree Algorithm in E-Learning System

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ABSTRACT
Learning style is an important factor that accounts for the individual student learning in any learning environment. Every student has a different learning style and different ways to percept, process, retain and understand new information. In this paper, a new approach is proposed to classify students learning style automatically and dynamically depending on their learning behavior in a learning management system (LMS). There have been several approaches proposed for automatic learning style detection. One of the widely accepted and frequently used classification techniques is the decision tree classifier. The decision tree classifier mainly depends on the construction of strong decision rules which are required to identify learning styles accurately. The lack of strong decision rules would lead to the misclassification of individual students learning style. Hence, this paper mainly focuses on the construction of strong decision rules to strengthen the existing decision tree classifier to accurately and precisely classify students learning style thereby improving the classification accuracy. The proposed approach has experimented with an average of 300 students enrolled for the online courses in Moodle LMS. Initially, the students' behavior are extracted from the web log files of LMS and then preprocessed to build decision tree classifier using strong decision rules based on the three learning dimensions of standard Felder Silverman learning style model (FSLSM). The evaluation result is obtained using the inference engine with forward reasoning searches of the rules until the correct learning style is determined. Based on the result obtained, the prediction of learning style is done for the new students automatically and accurately using the significant rules built in the decision tree classifier. The experimental result proved that the processing dimension shows variance in classification whereas perception and input dimension shows less variance with an average accuracy of 87%.

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DSIT2018, July 20–22, 2018, Singapore, Singapore
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ACM ISBN 978-1-4503-6521-5/18/07…$15.00
https://doi.org/10.1145/3239283.3239319

CCS Concepts
• Theory of computation→ Design and analysis of algorithms.

Keywords
Algorithm; Decision Tree classifier; Felder Silverman learning style model; Learning Style; Prediction; Weblog.

1. INTRODUCTION
Learning style is the best way the student wishes to learn. Each student has a different style of learning which helps them to achieve their full potential. The learning style of a learner determines how learners acquire knowledge effectively while learning. It is characterized by different types of factors such as course contents i.e., notes, exams, assignments, tutorials and exercises; the order in which course is organized and presented; how information is presented and communicated to the learners; the support of interaction among learners using online tools such as discussion forums, chat rooms, and email systems. The problem of identifying how learners would learn and attain new knowledge and skills has been gained the strong interest in research over the last decade. Hence there is a need to determine each individual learners learning style and assist the learner to best fit each learner needs.

The proposed approach is used to determine and predict the student learning styles on their learning performance in the learning system and also to evaluate the classification accuracy of the modified decision tree classifier with other classifiers.

2. LITERATURE REVIEW
There are several approaches used to acquire the learning style of students. The traditional method uses a static and time-consuming approach using questionnaires. This method is not accurate as the students are reluctant to answer the questions. With the introduction of LMS in education, automatic detection method has become more dynamic and accurate to acquire learning style. The two main automatic detection methods used are data-driven and literature-based. The data-driven method [1] uses Artificial Intelligence (AI) classification algorithms to determine the learning style, by taking learners model as an input. This method provides the more accurate result as it uses real student data and the error rate is less compared to the other methods. The literature-based methods determine learning style from a number of matching hint using simple rules. The drawback of this method is that

there is a possibility of missing some learning style behavioral patterns for the calculation of learning style [2].
The proposed system employs a data-driven approach using AI classification algorithm to classify student learning styles. Several learning styles models have been described such as Gardner, Kolb's, FSLSM, Biggs, and custom [1]. Each model has its own set of strengths and weaknesses. Among these models, FSLSM is chosen in the proposed approach as it is the widely used model in educational systems exhibiting a good degree of reliability, internal consistency, and validity. This model defines the learners learning style into four different dimensions (active/reflective, sensitive/intuitive, visual/verbal, sequential/global) based on the behavior patterns of learners using the system.

There were several research works carried out on a data driven approach using AI algorithms based on FSLSM. The various AI techniques used are Genetic algorithms with K-NN [3], Decision Tree [4], Neural Network [5], Bayesian network [6], Genetic Algorithms [7] etc.

As each algorithm has its own strengths and weaknesses, aiming at the most widely used AI classification algorithms are the Bayesian network, Neural Network, and Decision Tree. The Bayesian network uses conditional probabilities to represent the uncertainty of data, but it has no universally accepted standard to construct the network from data. Neural networks are capable of learning from sets of training examples which makes them suited for classification with high accuracy, but have the drawback of high computational cost, no theoretical rule defined to determine the optimal number of hidden neurons, complex to define the number of hidden layers, lack of descriptive power and hard to find out rules for both inputs and outputs. The decision tree is a widely used classification algorithm which has proved to have high accuracy. This algorithm requires significant classification rules for learning style detection. The previous studies reported in [4], [8] uses decision tree implementation which does not show significant rules for learning styles detection based on FSLSM. Hence, the construction of most significant rules would be the main focus of the proposed system to strengthen the existing decision tree classifier in order to make accurate learning style classification and prediction of learners and to emphasis on the automatic classification of learning styles when weblog files are given as input to the classifier which is not been done by other researchers. The main reason for selecting the decision tree classifier is because of the understandable and visible decision rules and simplicity etc.

The paper is organized as follows: Section 3 gives an overview of the proposed approach. Section 4 presents the result of the experiments conducted and discussion on the achievements and Section 5 provides the conclusion.

3. PROPOSED APPROACH

3.1 Proposed Architecture

Figure 1 shows the architecture of the proposed approach for extracting learning style from web log files. Initially, web log file of each learner is collected which contain the records of learners' behavior while using the system and the participation in using the course related tools such as live chat, forum discussion, quizzes and class assignments based on the three dimensions (processing, perception and input) of FSLSM learning style model. Then, data preprocessing is done to extract the learner actions based on the standard FSLSM model. This stage analyses the web log files based on the three dimensions of FSLSM model as explained below:

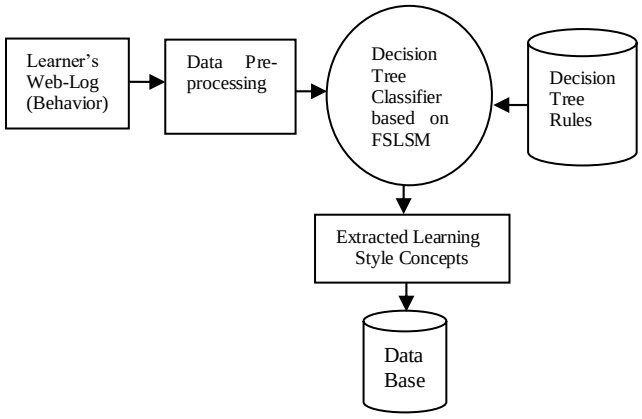


Figure 1. Architecture of extracting learning style.

Processing dimension: This dimension determines the active and reflective learners by their participation in doing activity something physical or theoretical. Active learners mainly prefer to work in groups and perform more exercises whereas reflective learners prefer to work alone and perform less number of exercises. Based on this learning behavior, the behavior patterns of learners such as forum discussions, assignments and chats are investigated from weblog files. For forum discussion, the actions analyzed are 'view & add post messages', 'add post to start discussions' 'only view forums & discussions' and 'no participation'. For chats, the actions analyzed are 'only view', 'view & report' and 'no participation' and for the assignments, 'view-submit assign form' and 'submit' actions are analyzed.

Perception dimension: This dimension determines the sensitive and intuitive learners. The sensitive learners mainly have more attentiveness, carefulness and low trial and high completion towards the performance in exams and exercises whereas intuitive learners' show being bored by details, carelessness and high trial and low completion in exams and exercises. Based on this learning behavior, the behavior patterns of learners such as assignments and exams are investigated from web log files. For exams, the actions analyzed are 'attempt and continue attempt', 'exam review', 'exams done' and 'exam delivery time' and for assignments, the actions analyzed are 'view-submit assign form' and 'submit' in order to determine the number of times the particular action is performed.

Input dimension: This dimension determines the visual and verbal learners by their preference over the learning content and the participation in group activities. Based on the characteristics of verbal and visual learners, behavior patterns such as powerpoint, text, and hyperlinks are investigated from web log files. For these behavior patterns, actions such as 'view' and 'no view' are analyzed to find the number of times the particular action is performed.

The extracted learners' actions are given as input to the modified decision tree classifier which includes more significant rules proposed as in Table 1, in order to strengthen the existing decision tree classifier to classify the learning style of each learner automatically, more precisely and accurately based on FSLSM. The final structure of a decision tree includes root, decision and leaf nodes with branches. Each node is labeled with the attributes it tests, and branches are labelled with possible values of the attribute. The process of classification involves mainly two phases:

the training and testing phase. The training phase constructs a perfect tree by partitioning the data set recursively into smaller subsets until all the records in a particular partition belong to the same class. In the testing phase, the attribute values of decision nodes are tested with test data set and iteratively pruned to obtain a tree with higher accuracy. Figure 3 shows the modeling of learner's behavior in Moodle LMS based on FSLSM in which internal nodes represent the FSLSM dimensions and the actions of learner behavior's accessing the system. The final step is to predict the learning style of the learners based on the input behavior of learners. Table 1 shows the categories and the rules that are to be applied to the input characteristics to do the prediction by the decision tree classifier model. Based on the rules applied to these actions, final values are calculated (Table 1) which is then used to calculate the final category of dimensions.

The algorithm of learning style prediction is shown in Figure 2.

Algorithm

Input: Weblog files of each learner

Output: Learning style of each learner in three dimensions of FSLSM

1. Access the web log files of each learner.
2. Preprocess weblog files
   Filter the activities and their corresponding actions of each learner from weblog files based on the three dimensions of FSLSM model.
3. Create learning style model
   a. Determine the activities of learners in Moodle LMS
   b. Build Tree using decision tree classifier
      i. Initially, Tree T is empty
      ii. T = Create root node and its splitting attribute
      iii. T = Add branch to the root node for each split attribute and label it with the test value
      For each branch do
      Move to the split attribute
      Test on the possible value of the split attribute
      Create a new branch below root based on the test result
      If stopping point is reached for this branch, create leaf a node with an appropriate label.
4. Predict learning style model
   a. Input is the learner activities
   b. Start at the root node
   c. Move to the decision node
   d. Test on each possible value of the decision node and descend the appropriate branch
   e. Test repeated until a leaf node is reached
      Return the classification of leaf node which is the learning style.

Figure 2. Algorithm of extracting learning style.

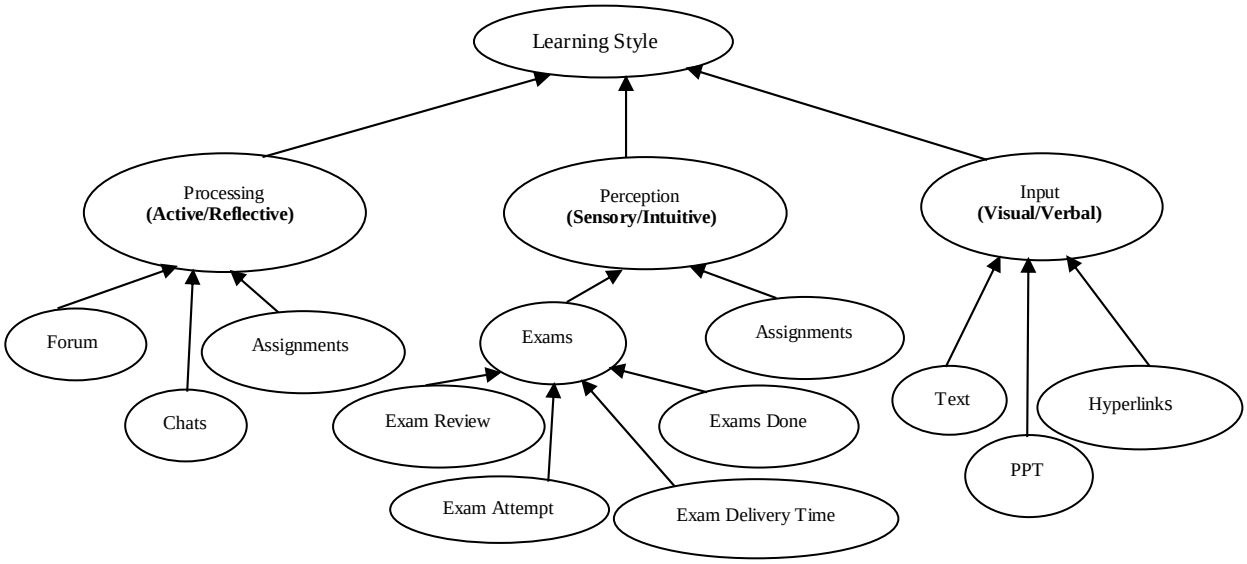


Figure 3. Modeling student's behavior with Moodle LMS.

Table 1. Decision tree rules

Dimensions	Category		Rules	Values (V)
Processing (Active /Reflective)	Forum	No participation	No participation	0
		Only view forums & discussions	Number of views	1
		View & add post	Number of view and post	2
		Add post for discussion	Number of post	3
	Chats	No participation	No participation	0
		Only view	Number of view	1
		View & report	Number of view and report	2
	Assignments Done	Submit, View-Submit assign form	Attend > 70% assignments	4
			Attend 50% - 70% assignments	3
			Attend 30% - 50% assignments	2
			Attend 10% - 30% assignments	1
			Attend <10% assignments	0
Perception (Sensory/ Intuitive)	Exams	Exam Review	Makes > 20% review	4
			Makes 15% - 20% review	3
			Makes 10% - 15% review	2
			Makes 5% - 10% review	1
			Makes <5% review	0
		Attempt & Continue Attempt	Attempt > 70% of exam	4
			Attempt 50% - 70% of exam	3
			Attempt 30% -50% of exam	2
			Attempt 10% - 30% of exam	1
			Attempt <10% of exam	0
		Exam Delivery Time	Utilize > 70% of time	4
			Utilize 60% - 70% of time	3
			Utilize 50% - 60% of time	2
			Utilize 40% - 50% of time	1
			Utilize <40% of time	0
		Exams done	Attend > 70% exams	4
			Attend 50% - 70% exams	3
			Attend 30% - 50% exams	2
			Attend 10% - 30% exams	1
			Attend <10% exams	0
	Assignments Done	Submit, View-Submit assign form	Attend > 70% of assignments	4
			Attend 50% - 70% assignments	3
			Attend 30% - 50% assignments	2
			Attend 10% - 30% assignments	1
			Attend <10% assignments	0
Input (Visual/ Verbal)	Text (Word)	View Course (Word)	Number of no view (Word)	0
		No view	Number of view (Word)	1
	Power point	View Course (Power point)	Number of no view(powerpoint)	0
		No view	Number of view (powerpoint)	1
	Hyperlinks	View Course (HyperLink)	Number of no view (HyperLink)	0
		No view	Number of view (Hyper Link)	1

V	Processing Dimension Values
8-9	Extremely Active
6-7	Medium Active
4-5	Neutral
2-3	Medium Reflective
0-1	Extremely Reflective

V	Perception Dimension Values
16<S<=20	Extremely Sensitive
12<S<=16	Medium Sensitive
8<S<=12	Neutral
4<S<=8	Medium Intuitive
0<=S<=4	Extremely Intuitive

V	Input Dimension Values
3	Extremely Visual
2	Neutral
1	Extremely Verbal
0	-

4. RESULTS AND DISCUSSION

The proposed approach has experimented on the real-time web log files of approximately around 300 datasets of learners collected from Moodle LMS used in the institution over a period of one year studying across 3 semesters for 5 online computing courses of computing department. The performance of the proposed decision tree classifier with significant rules is evaluated with other standard classifiers using various performance metrics in order to determine the efficiency of the proposed classifier.

Confusion Matrix: The confusion matrix is one of the metrics which is used to illustrate the accuracy of the solution in a classification problem. It contains information about both actual and predicted classifications of the system. The performance of such a system is evaluated using the data in the matrix. Table 2 shows the confusion matrix obtained for the three dimensions of FSLSM.

Table 2. Confusion matrix of three dimensions of FSLSM

Perception Predicted /Actual	**Extremely Intuitive**	**Medium Intuitive**	**Neutral**	**Medium Sensitive**	**Extremely Sensitive**
Extremely_ Intuitive	2	0	0	0	0
Medium_ Intuitive	0	1	0	0	0
Neutral	0	0	150	0	0
Medium_ Sensitive	0	0	0	56	0
Extremely_ Sensitive	0	0	0	0	1

Processing Predicted /Actual	**Extremely Reflective**	**Medium Reflective**	**Neutral**	**Medium Active**	**Extremely Active**
Extremely_ Reflective	6	0	0	0	0
Medium_ Reflective	0	12	0	0	0
Neutral	0	0	150	0	0
Medium_ Active	0	0	0	39	0
Extremely_ Active	0	0	0	0	3

Input Predicted /Actual	**Neutral**	**Extremely Visual**	**Extremely Verbal**
Neutral	190	0	0
Extremely_Visual	0	13	0
Extremely_Verbal	0	0	7

Precision and Recall: The confusion matrix is then used to compute the precision and recall for each class. Precision is the fraction of relevant instances among retrieved instances, while recall is the fraction of relevant instances that have been retrieved as in Eq. (1).

Precision = True Positive / Total Predicted

Recall = True Positive / Total Retrieved - Eq. (1)

The values of precision and recall calculated to three dimensions of FSLSM learning styles are compared with the metrics of standard classifiers, J48, Hoeffding, Random and Logic Model Tree obtained from WEKA program using 10 fold cross validation method as shown in Figure 4. According to the values defined in the tables, the precision of the proposed decision classification algorithm is high compared to other classification technique.

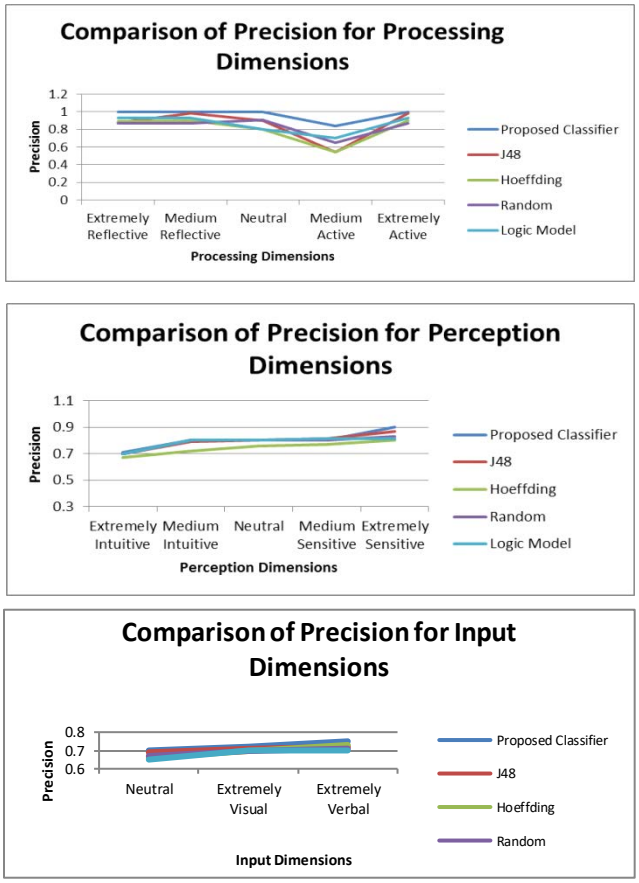


Figure 4. Comparison of precision values for FSLSM dimensions.

Classification Accuracy: The performance is also done by evaluating classification accuracy which determines the percentage of correctly classified instances placed in the correct category. The Eq. (2) gives the formula of classification accuracy and error rate:

Accuracy = Correct Predictions / Total Number of Predictions

Error Rate = Wrong Predictions / Total Number of Predictions - Eq. (2)

The achieved results for classification accuracy and error rate obtained for proposed decision tree classifier in Figure 5 ranges from 88% to 95% shows a high accuracy for the three dimensions of FSLSM compared to other standard classifiers, thereby proving

its suitability for identifying the learning style of learners with an average accuracy of 91% and a very less error rate of 9%.

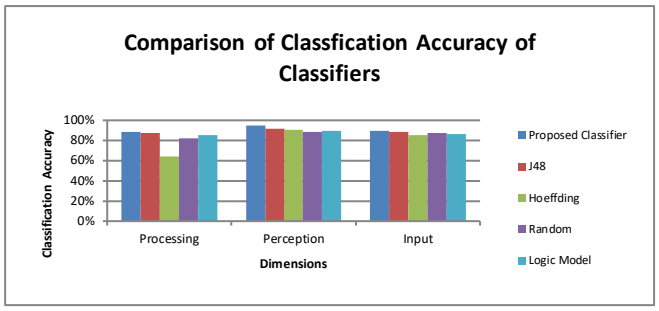


Figure 5. Comparison of classification accuracy of various classifiers.

Figure 5 shows the comparison of classification accuracy of various classifiers. By analyzing the evaluation results obtained, the proposed decision tree classifier is able to classify perception dimension with a high accuracy of 95% as most learners did most of the exams and exercises provided for the course and hence significant variance was not found in this dimension. The input dimension is classified with a precision of 90% since most of the students use power point and hyperlinks rather than theoretical notes and hence less variance is shown in the predicted class. The processing dimension is classified with a precision of 88% as the students use forums and chats very less and hence variance was found in this dimension. The variation could be improved if the students were encouraged to use the facilities of LMS such as forums, chat rooms, email etc. effectively and also to use collaborative tasks such as assignments and exercises.

From the results of the various experiments conducted, it is proved that the significant rules of proposed decision tree is able to classify students learning style accurately with high precision and accuracy provided that students are encouraged to use the group facilities effectively in Moodle LMS and also it is evident that learning style obtained are consistent with the learners preferences based on FSLSM model.

5. CONCLUSION

The proposed approach predicts the learning preference of students while using the Moodle LMS based on three learning dimensions of FSLSM. The approach uses the modified decision tree classifier built with strong decision rules to accurately classify the learning style based on FSLSM dimension: processing, perception and input. The final result proved that the modified decision tree classifier is able to detect and predict learning style with an accuracy of 75% in the processing dimension, 95% in the perception dimension, and 90% in the input dimension thereby obtaining an overall high accuracy of 87% compared with other standard classifiers. The final result would help educators to assist in framing learning course materials based on the students learning style.

The future research is to design an effective adaptive system by integrating the classifier into LMS for automatic student learning style detection.

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